MAANAS TYAGI

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PROFESSIONAL SUMMARY

AI/ML engineer with hands-on experience building production AI systems, automation pipelines, and evaluated retrieval-augmented generation (RAG) pipelines. Proficient in machine learning, deep learning, NLP, and multi-agent agentic AI architectures. Author of one preprint and one paper currently under review on large language model (LLM) behavior.

SELECTED IMPACT

- Demonstrated through a 33-query, RAGAS-scored ablation study that semantic-only retrieval outperforms hybrid BM25+semantic retrieval on every metric (Recall@5: 0.848 vs. 0.818, MRR: 0.703 vs. 0.610), overturning the assumed hybrid-retrieval default
- Replaced a $3,000 licensed reconciliation tool with a custom Python automation pipeline at Genpact, reducing 10,000-row file processing time from over 30 minutes to under 5 minutes for Fortune 500 clients; now actively used by tens of colleagues to automate their daily reconciliation work
- Contributed 5 merged pull requests to Hugging Face repositories (tau: #314, #290, #289; Transformers: #45150), each reviewed and accepted by project maintainers
- Designed and built end-to-end AI systems at Sstudize, including system architecture and the application layer for Student GPT, along with a token economics model for sustainable per-user LLM cost structure
- Co-built TrenTorch to 130+ GitHub stars in 2 days of launch — one of the fastest-growing open-source educational ML repositories on GitHub, with 10+ forks

PROFESSIONAL EXPERIENCE

LLM Engineer *July 2026 – Present*
Sstudize · Remote

- Architect and own all AI systems for the department, including system design and the application layer for Student GPT, an AI-powered platform for students
- Developed a token economics model to define sustainable per-user LLM cost and usage structure for the product

AI & Analytics Intern (*Forward Deployed Engineer Level Work*) *May 2026 – June 2026*
Genpact · Jaipur, India

- Replaced a $3,000 licensed reconciliation tool with a custom Python automation pipeline, reducing processing time for 10,000-row files from over 30 minutes to under 5 minutes; now actively used by tens of colleagues for daily reconciliation work
- Built purchase, invoice, and GRNI reconciliation engines for Fortune 500 clients, converting manual pivot-table workflows into single-file uploads with instant, fully cross-matched output

PUBLICATIONS

- **Instruction Fidelity Decay in Small Language Model Agents: Semantic Recall Dissociates from Format Compliance Under Semantic Interference** (Under Review)
- **Front-Loading Bias in Large Language Models: A Multi-Model Empirical Analysis** (ResearchGate Preprint) – doi.org/10.13140/RG.2.2.32504.92164

PROJECTS

RAG Eval System | *LangChain · Chroma · RAGAS · Python* *July 2026*
github.com/maanas1234/RAG-Eval-System

- Constructed a retrieval and generation evaluation harness using Precision/Recall/MRR and RAGAS faithfulness/relevancy metrics, tested against 33 hand-labeled queries over a 69-document corpus, scoring retrieval and generation quality independently
- Ablation study: semantic-only retrieval outperformed hybrid BM25+semantic retrieval on every metric (Recall@5: 0.848 vs. 0.818, Faithfulness: 0.788 vs. hybrid), overturning the assumed hybrid-is-better default

Personal Intelligence System | *Python · FastAPI · OpenRouter · Multi-Agent* *Ongoing*

- Built a self-hosted personal knowledge engine with 13 plug-and-play connectors (Notion, Slack, Email/IMAP, GitHub, Jira, Linear, Confluence, and more) that chunk, embed, and store all ingested content in a semantic vector space queryable in natural language
- Architected a two-layer agent system: Boss Agent auto-schedules syncs across all connected sources; Brain Agent dispatches to per-source sub-agents with retry/backoff and hash-based content dedup to skip unchanged items

SLM Pretraining From Scratch | *PyTorch · Python* *June – July 2026*

- Implemented a decoder-only transformer (GPT-style architecture) from scratch, including attention mechanism, tokenizer, and training loop, without pretrained weights or external frameworks; 2.1M parameters
- Trained to a validation loss of 1.74 (~5.7 perplexity) using cosine learning rate decay with warmup; evaluated 5 architecture configurations from 120K to 4.8M parameters to characterize the relationship between model size and overfitting in a low-data regime

TrenTorch | *Python · NumPy · OSS* *Aug 2026 – Present*

- Core engineer on TrenTorch — 40+ merged PRs across the tren CLI (github.com/TrenTorch/TrenTorch) and TrenTorch-Web IDE; own the CLI end-to-end: test runner correctness, Jupyter server integration, benchmarking pipeline, progress tracking, Windows compatibility
- Authored 25+ curriculum modules on TrenTorch-Web: Systems Performance (quantization, mixed precision, CUDA kernels, profiling), Post-Training Alignment (RLHF, DPO, GRPO, QLoRA), Deployment & Serving (DAGs, canary, A/B testing), and Drift Monitoring (PSI, concept drift detection)

EDUCATION

Vellore Institute of Technology *Vellore, India*
Bachelor of Technology in Information Technology, 2024 – 2028

SKILLS

Languages: Python, SQL, C
RAG / Retrieval: LangChain, RAGAS, Chroma, FAISS, Embeddings, Vector Databases, Semantic Search, CRAG
Modeling: PyTorch, Machine Learning, Deep Learning, Transformer Architecture, Fine-Tuning (LoRA/PEFT), Prompt Engineering
Backend / Tools: Hugging Face, FastAPI, Flask, React, Pandas, NumPy, Docker, Git, PostgreSQL, MySQL